Lecture 17 (153–248)

This week: AR
Next week: MA models & then ARMA + ARIMA

Vector time series
Neural Net

Auto Regression (AR)

$\{y_t\}$

$\varepsilon_t \overset{i.i.d}{\sim} N(0, \sigma^2)$

$$y_t = \phi_0 + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

Usual regression of y_t on its lagged/past values: $y_{t-1}, \ldots, y_{t-p}$

$$y = \begin{pmatrix} y_{p+1} \\ \vdots \\ y_n \end{pmatrix} \qquad X = \begin{bmatrix} 1 & y_{t-1} & y_{t-2} & \cdots & y_{t-p} \\ \vdots & & & & \\ 1 & & & & \end{bmatrix}$$

time $t = p+1, \ldots, n$

$t = p+1, \ldots, n$

sm.OLS(y, X).

For simplicity, $p = 1$ (AR(1))

$$y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t$$

Usual Regression

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

Data: $(x_1, y_1) \dots (x_m, y_m)$

Likelihood: β_0, β_1, σ

$$f_{(x_1, y_1) \dots (x_m, y_m) \mid \beta_0, \beta_1, \sigma}(x_1, y_1, x_2, y_2, \dots, x_m, y_m)$$

$$= f_{y_1 \dots y_m \mid x_1 \dots x_m, \beta_0, \beta_1, \sigma}(y_1 \dots y_m) \cdot f_{x_1 \dots x_m \mid \beta_0, \beta_1, \sigma}(x_1 \dots x_m)$$

Assume this away.

$$\propto f_{y_1 \dots y_m \mid x_1 \dots x_m, \beta_0, \beta_1, \sigma}(y_1 \dots y_m)$$

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$= f_{\beta_0 + \beta_1 x_1 + \varepsilon_1, \dots \beta_0 + \beta_1 x_m + \varepsilon_m \mid x_1 \dots x_m, \beta_0, \beta_1, \sigma}(y_1 \dots y_m)$$

$$= f_{\varepsilon_1 \dots \varepsilon_m \mid x_1 \dots x_m, \beta_0, \beta_1, \sigma}(y_1 - \beta_0 - \beta_1 x_1, \dots, y_m - \beta_0 - \beta_1 x_m)$$

Assume $\varepsilon_1 \dots \varepsilon_m$ are independent of $x_1 \dots x_m$ (for fixed β_0, β_1, σ)

$$\rightarrow f_{\varepsilon_1 \cdots \varepsilon_m \mid \beta_0, \beta_1, \sigma}(y_1 - \beta_0 - \beta_1 x_1, \ldots, y_m - \beta_0 - \beta_1 x_m)$$

$$\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$$

$$\propto \sigma^{-m} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^{m} (y_i - \beta_0 - \beta_1 x_i)^2\right]$$

Assumptions:

① $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

② $x_1 \ldots x_m \mid \beta_0, \beta_1, \sigma$ does not depend on the parameters

③ $\varepsilon_1 \ldots \varepsilon_m$ are independent of $x_1 \ldots x_m$ (given β_0, β_1, σ)

④ $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

AR (AUTO REGRESSION)

$i = t \quad (y_1, y_2), (y_2, y_3), \ldots, (y_{n-1}, y_n)$

$m = n - 1$

$x_t = y_{t-1}$

$$y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t, \quad t = 2, \ldots, n$$

$\theta = (\beta_0, \beta_1, \sigma)$

Data: $y_1 \ldots y_n$

Likelihood $= f_{y_1 \ldots y_n \mid \beta_0, \beta_1, \sigma}(y_1 \ldots y_n)$

$$= f_{y_1 \mid \beta_0, \beta_1, \sigma}(y_1) \; f_{y_2 \mid y_1, \theta}(y_2) \; f_{y_3 \mid y_1, y_2, \theta}(y_3) \cdots f_{y_n \mid y_1 \ldots y_{n-1}, \theta}(y_n)$$

$$= f_{y_1|\theta}(y_1)\, f_{\phi_0+\phi_1 y_1+\varepsilon_2 \mid y_1,\theta}(y_2)\, f_{\phi_0+\phi_1 y_2+\varepsilon_3 \mid y_1,y_2,\theta}(y_3) \cdots f_{\phi_0+\phi_1 y_{t-1}+\varepsilon_t \mid y_1 \cdots y_{t-1},\theta}(y_t) \cdots f_{\phi_0+\phi_1 y_{n-1}+\varepsilon_n \mid y_1 \cdots y_{n-1},\theta}(y_n)$$

$$= f_{y_1|\theta}(y_1) \prod_{t=2}^{n} f_{\varepsilon_t \mid y_1 \cdots y_{t-1},\theta}(y_t - \phi_0 - \phi_1 y_{t-1})$$

Assumption: ε_t is independent of past y's $y_1 \cdots y_{t-1}$ (for fixed ϕ_0, ϕ_1, σ) for each t

$$= f_{y_1|\theta}(y_1) \prod_{t=2}^{n} f_{\varepsilon_t|\theta}(y_t - \phi_0 - \phi_1 y_{t-1})$$

ε_t iid $N(0,\sigma^2)$

$$\propto f_{y_1|\theta}(y_1) \prod_{t=2}^{n} \frac{1}{\sigma} \exp\left(-\frac{1}{2\sigma^2}(y_t - \phi_0 - \phi_1 y_{t-1})^2\right)$$

$$= f_{y_1|\theta}(y_1)\, \sigma^{-(n-1)} \exp\left[-\frac{\sum_{t=2}^{n}(y_t - \phi_0 - \phi_1 y_{t-1})^2}{2\sigma^2}\right]$$

SAME Likelihood as in usual regression

Assumption: ① $y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t$

② ε_t is independent of <u>past</u> y_t's

$y_{t-1}, y_{t-2}, \ldots, y_1$

③ $\varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$

How to deal with $f_{y_1|\theta}(y_1)$?

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① Ignore $f_{y_1|\theta}(y_1)$ [CONDITIONAL LIKELIHOOD]

$f_{y_2 \cdots y_n | y_1, \theta}(y_2 \cdots y_n)$

② Derive $f_{y_1|\theta}(y_1)$

$$y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t, \quad t = 0, 1, 2, \ldots, n$$

$$y_1 = \phi_0 + \phi_1 y_0 + \varepsilon_1$$

$$= \phi_0 + \phi_1 (\phi_0 + \phi_1 y_{-1} + \varepsilon_0) + \varepsilon_1$$

$$= \phi_0 + \phi_1 \phi_0 + \phi_1^2 y_{-1} + \phi_1 \varepsilon_0 + \varepsilon_1$$

$$= \phi_0 + \phi_1 \phi_0 + \phi_1^2 (\phi_0 + \phi_1 y_{-2} + \varepsilon_{-1}) + \phi_1 \varepsilon_0 + \varepsilon_1$$

$$= \phi_0 + \phi_1 \phi_0 + \phi_1^2 \phi_0 + \phi_1^3 y_{-2} + \phi_1^2 \varepsilon_{-1} + \phi_1 \varepsilon_0 + \varepsilon_1$$

$$= \text{REPEAT}$$

$$= \phi_0 (1 + \phi_1 + \phi_1^2 + \cdots + \phi_1^M) + \phi_1^{M+1} y_{-M} + \varepsilon_1 + \varepsilon_0 \phi_1 + \varepsilon_{-1} \phi_1^2 + \cdots + \phi_1^M \varepsilon_{1-M}$$

If $|\phi_1| < 1$,

$M \to \infty$ ⟶ $\phi_0 (1 + \phi_1 + \phi_1^2 + \cdots) + \varepsilon_1 + \varepsilon_0 \phi_1 + \varepsilon_{-1} \phi_1^2 + \varepsilon_{-2} \phi_1^3 + \cdots$

$$= \frac{\phi_0}{1-\phi_1} + \varepsilon_1 + \varepsilon_0\phi_1 + \varepsilon_{-1}\phi_1^2 + \varepsilon_{-2}\phi_1^3 + \cdots$$

$$y_1 \sim N\left(\frac{\phi_0}{1-\phi_1}, \left(1+\phi_1^2+(\phi_1^2)^2+(\phi_1^3)^2+\cdots\right)\sigma^2\right)$$

$$= N\left(\frac{\phi_0}{1-\phi_1}, \frac{\sigma^2}{1-\phi_1^2}\right) \longrightarrow \text{ASSUME } |\phi_1|<1$$

$$f(y_1) \propto \frac{\sqrt{1-\phi_1^2}}{\sigma} \exp\left[\frac{-\left(y_1-\frac{\phi_0}{1-\phi_1}\right)^2(1-\phi_1^2)}{2\sigma^2}\right]$$

$y_1 \mid \theta$

AR(1) $\longleftrightarrow$ $|\phi_1|<1$

Frequentist vs Bayesian for AR

AR(1) Frequentist

Fix y_1.

ε_t independent of $y_\cdots y_{t-1}$ & ε_t i.i.d $N(0,\sigma^2)$

Conditional likelihood (given y_1)

$$\sigma^{-(n-1)} \exp\left[\frac{-1}{2\sigma^2}\sum(y_t-\phi_0-\phi_1 y_{t-1})^2\right]$$

$$\hat{\phi}_1 = \frac{\sum(y_t-\bar{y})(y_{t-1}-\bar{x})}{\sum(y_{t-1}-\bar{x})^2}$$

$\bar{y} = \frac{y_2+\cdots+y_n}{n-1}$

$$\hat{\phi}_1 = \frac{\sum_{t=2}^{n} \left[y_t - \frac{y_2 + \cdots + y_n}{n-1} \right] \left[y_{t-1} - \frac{y_1 + \cdots + y_{n-1}}{n-1} \right]}{\sum_{t=2}^{n} \left(y_{t-1} - \frac{y_1 + \cdots + y_{n-1}}{n-1} \right)^2}$$

where $\bar{y} = \frac{y_2 + \cdots + y_n}{n-1}$ and $\bar{x} = \frac{y_1 + \cdots + y_{n-1}}{n-1}$

→ For frequentist inference, the distribution of $\hat{\phi}_1$ needs to be computed under AR(1) assumptions.

$$\frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$$

Bayesian inference

Likelihood: $\sigma^{-(n-1)} \exp\left(-\frac{\sum_{t=2}^{n} (y_t - \phi_0 - \phi_1 y_{t-1})^2}{2\sigma^2} \right)$

prior: $\phi_0, \phi_1, \log\sigma$ i.i.d Unif$(-c, c)$, $c \to \infty$

Usual t-inference: Sm.OLS(Y, X)

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

AR(p)

$$y_t = \phi_0 + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

Regression can be justified by conditional likelihood (conditional density of $y_{p+1}, \ldots, y_n$ given $y_1 \ldots y_p$)

$$\sigma^{-(n-p)} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{t=p+1}^{n} \left(y_t - \phi_0 - \phi_1 y_{t-1} - \cdots - \phi_p y_{t-p} \right)^2 \right\}$$

Predictions

$\hat{\phi}_0, \hat{\phi}_1$

AR(1):

$$\hat{y}_{n+1} = \hat{\phi}_0 + \hat{\phi}_1 y_n$$

$$\hat{y}_{n+2} = \hat{\phi}_0 + \hat{\phi}_1 \hat{y}_{n+1}$$

$$= \hat{\phi}_0 + \hat{\phi}_1 (\hat{\phi}_0 + \hat{\phi}_1 y_n)$$

$$= \hat{\phi}_0 + \hat{\phi}_1 \hat{\phi}_0 + \hat{\phi}_1^2 y_n$$

$$\hat{y}_{n+3} = \hat{\phi}_0 + \hat{\phi}_1 \hat{\phi}_0 + \hat{\phi}_1^2 \hat{\phi}_0 + \hat{\phi}_1^3 y_n$$

$$\hat{y}_{n+k} = \hat{\phi}_0 \left(1 + \hat{\phi}_1 + \cdots + \hat{\phi}_1^{k-1} \right) + \hat{\phi}_1^k y_n$$

If $|\hat{\phi}_1| > 1$

$\hat{\phi}_1 = 1$

$\hat{y}_{n+k} = k\hat{\phi}_0 + y_n$ → linear prediction

$|\hat{\phi}_1| < 1$ → $\hat{y}_{n+k}$ → constant

AR(1)

$\hat{\phi}_1 = 1$

$y_t = \hat{\phi}_0 + \hat{\phi}_1 y_{t-1} + \varepsilon_t$

$y_t = \hat{\phi}_0 + \varepsilon_t$

$y_t - y_{t-1}$

$\hat{\phi}_1$

1.003